

1 Quantum Error Correction

The main challenge of quantum computation is the fragility of the physical systems that are used to implement the qubits. In classical computation, information is encoded in macroscopic physical variables, and the logical states are separated by energy barriers that make them robust against environmental fluctuations. Compared to classical bits, qubits are heavily affected by environmental noise, which makes them degrade and lose their information due to decoherence. It is thus necessary to explore the most efficient ways to protect quantum information while maintaining the practical feasibility of operating with those protected qubits.

1.1 Overview and Simple Codes

Fathoming the differences between classical and quantum error-correcting codes makes it necessary to pay attention to three difficulties that arise when we deal with quantum systems: The no-cloning theorem, which prevents us from duplicating the state of a qubit; the continuous nature of errors; and the fact that observation in quantum mechanics collapses the wave function in one of the eigenstates of the measured observable (this is a non-unitary process, and thus recovery is impossible)

In order to overcome these difficulties, quantum error-correcting codes (QECC) utilize entanglement to encode a logical qubit across a multi-qubit system. This way, we can protect the quantum information via redundancy, without violating the no-cloning theorem. QEC protocols measure multi-qubit parity operators (error syndromes), avoiding the collapse of the wave function. The continuity problem will be confronted by discretizing the errors into a finite set of Pauli operations.

We will build some basic intuition about quantum error correction models based on redundancy by looking at one of its earliest and simplest examples: the Shor Nine-Qubit Code.

Consider that we have physical qubits that are subject to random bit-flip errors, that is, with a certain probability p , the state $|\psi\rangle$ is transformed into $X|\psi\rangle$. As we cannot simply duplicate our original qubit, we can entangle it with two ancilla qubits initialized in the $|0\rangle$ state by applying two CNOT gates, where the original qubit is the control and the ancilla qubits are the targets. If our original state is $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the resulting state will be $|\psi_L\rangle = \alpha|000\rangle + \beta|111\rangle$. The encoding establishes a logical basis with states:

$$\begin{aligned} |0_L\rangle &= |000\rangle \\ |1_L\rangle &= |111\rangle \end{aligned}$$

If the bit-flip error occurs in any qubit, we can diagnose this error without destroying the superposition by measuring the parity observables Z_1Z_2 and Z_2Z_3 . Those observables compare the parities of their corresponding qubits, yielding a ± 1 eigenvalue, positive if both compared qubits share the same state, negative if they are different. It is crucial to note that the measurement of the parity operators commute with the logical operators, ensuring that the superposition remains intact. Once we have detected the error, we can

apply the appropriate recovery operation.

This same procedure can be adapted to protect against phase-flip errors, modeled by the Pauli Z operator. This is performed by applying a Hadamard transformation to each qubit after applying two sequential CNOT gates. It is an identical procedure, and it only differs in the basis we are working with.

If we concatenate the bit-flip and phase-flip codes, we can protect against an arbitrary error on a single qubit. This is therefore a nine qubit code, with the bases states defined as follows:

$$|0_L\rangle = \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)$$

$$|1_L\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)$$

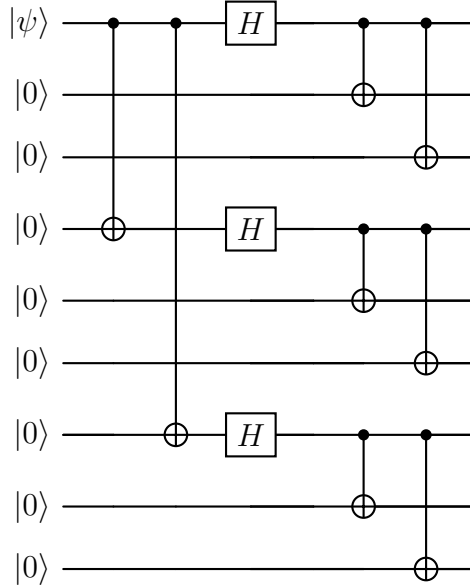


Figure 1: Shor Code circuit

The syndrome measurement is performed in an analogous way. We first compare the first two qubits by applying the operator Z_1Z_2 , then we continue comparing each pair of neighboring qubits by applying the corresponding operators. We can detect the bit-flip errors on any of the qubit. In order to detect phase-flip errors, we compare the signs of each of the three-qubits blocks.

The Shor code allows us to detect bit-flip and phase-flip errors. In order to illustrate that this condition is enough to detect an arbitrary error, we will describe the noise by a trace-preserving operation $\mathcal{E}$. Given ρ the density matrix of the system, the noise acting on the state can be expressed as:

$$\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger \quad (1)$$

This is called Kraus operator-sum representation, and will be discussed in more detail. The operators E_k can be expanded as a linear combination of the Pauli matrices $E_k = e_{i0}I + e_{i1}X_1 + e_{i2}Z_1 + e_{i3}X_1Z_1$. The resulting quantum state $E_k|\psi\rangle$ can be expressed as a superposition of $|\psi\rangle$, $X_1|\psi\rangle$, $Z_1|\psi\rangle$, and $X_1Z_1|\psi\rangle$. When we measure the error syndrome, the system collapses to any of the four states, and after applying the necessary correction operation, we obtain the final state $|\Psi\rangle$.

1.2 General Theory of Quantum Error Correction

Once we have built some intuition about QEC, we can construct a more general theory.

Firstly, we need to give a mathematical description of our noise $\mathcal{E}$. Physically, noise represents an interaction between the system and the environment. Since we do not have access to the environment, the action of noise on the system appears to be a non-unitary operation, and we cannot describe our qubit as if it was in a pure state. This forces us to use the density matrix

$$\rho = \sum_i p_i |\psi_i\rangle \langle \psi_i| \quad (2)$$

that represents the mixture of different pure states $|\psi_i\rangle$ with probabilities p_i . This must be a semi-positive-definite Hermitian matrix, with unit trace and non-negative eigenvalues. Non-unitary noise operations must preserve these properties. Hence, we represent noise acting on a system as a transformation of the state described by the density that fulfill this condition: a Completely Positive Trace Preserving (CPTP) map. In the Kraus representation, the action of a CPTP map $\mathcal{E}$ on a density matrix ρ is given by the formula 1, where the Kraus operators E_k satisfy the condition $\sum_k E_k^\dagger E_k = I$ (this condition derives necessarily from the requirement of trace preservation). If we consider any state $|\psi\rangle$ in the Hilbert space and consider the set of error-affected states $|\psi_k\rangle = E_k^\dagger |\psi\rangle$, we can easily check that:

$$\sum_k \langle \psi | E_k \rho E_k^\dagger | \psi \rangle = \sum_k \langle \psi_k | \rho | \psi_k \rangle \geq 0$$

which verifies that the map is completely positive. We will call the operation elements $\{E_i\}$ errors, and they represent the transformation of the quantum state under Hamiltonian parameter deviations from their ideal values.

Now we want to generalize the basic ideas present in the Shor code to give a simple picture of how we can detect and correct errors in a quantum system. We firstly define a quantum error-correcting code as a subspace C of the total Hilbert space, and the elements of C will be called codewords. We encode our quantum states by projecting them into C by an unitary operation P that we will call the projector. The encoded system is then vulnerable to environmental noise, which we model as a CPTP map $\mathcal{E}$. We perform a syndrome measurement to determine the error that has affected the system, and a recovery operation $\mathcal{R}$ returns the system to its original encoded state.

When we project our state into the code subspace, we are essentially restricting the degrees of freedom of the system. This means that if our original Hilbert space for n physical qubits is 2^n -dimensional, the code subspace will be 2^k -dimensional, where $k < n$. We are encoding k logical qubits into n physical qubits. Once the noise acts on the encoded state $|\psi_L\rangle$, the system will be pushed into a different subspace. Specifically, each error E_α projects the state into an error subspace $C_\alpha = E_\alpha C$. Each error subspace has its corresponding projector $P_\alpha = E_\alpha P E_\alpha^\dagger$.

If, given a code, each error corresponds to different undeformed and orthogonal subspaces, then we can reliably distinguish which error has occurred. This will be achieved by performing the set of projective measurements associated with each error subspace. Given our noisy state $E_\alpha |\psi_L\rangle$, the execution of a projector P_β will yield us a syndrome lecture β with probability $p_\beta = \langle \psi_L | E_\alpha^\dagger P_\beta E_\alpha | \psi_L \rangle$. The orthogonality of error subspaces ensures that the measurement associated to an error subspace C_β will only yield a non-zero probability when the corresponding error has occurred.

The undeformed condition means that, if you have two perfectly distinguishable (orthogonal) states $|\psi_L\rangle$ and $|\phi_L\rangle$ in the code subspace, then the error E_α must not make them non orthogonal. This is equivalent to saying that the inner product of the two states must be preserved under the action of the error, that is, $\langle \psi_L | E_\alpha^\dagger E_\alpha | \phi_L \rangle = \langle \psi_L | \phi_L \rangle = 0$. This ensures that the recovery operation can correctly restore the original encoded state.

We can define the condition for an error correction code to be able to correct the effect of noise as follows: Given a code C , a noisy quantum channel characterized by $\mathcal{E}$, and any state $|\psi_L\rangle$, with density matrix ρ_L , the error correction is successful if

$$(\mathcal{R} \circ \mathcal{E})(\rho) \propto \rho \quad (3)$$

In this condition, and in the following equation, we have not imposed the trace-preserving condition for $\mathcal{E}$, with the idea of allowing for more general noise models. That is the reason we use the proportionality symbol $\propto$. The prior discussion can be summarized in a very compact algebraic expression, the Knill-Laflamme condition for quantum error correction:

Theorem 1 (Knill-Laflamme). *Let C be a quantum error-correcting code, P the projector onto C , and $\mathcal{E}$ a CPTP map with operation elements $\{E_i\}$. Then, there exists a recovery operation $\mathcal{R}$ such that $(\mathcal{R} \circ \mathcal{E})(\rho) = \rho$ for all ρ supported on C if and only if the following condition holds:*

$$P E_i^\dagger E_j P = \alpha_{ij} P \quad (4)$$

for some Hermitian matrix α_{ij} .

Proof. ($\Leftarrow$) We notice that the Hermitian matrix α_{ij} can be diagonalized by a unitary transformation u , such that $u \alpha u^\dagger = d$, where d is a diagonal matrix. We can define new operation elements $F_i = \sum_j u_{ij} E_j$. Those can be proven to be operation elements of $\mathcal{E}$.

The new operation elements also satisfy the Knill-Laflamme condition, since:

$$\begin{aligned}
PF_i^\dagger F_j P &= \sum_{k,l} u_{ki}^\dagger u_{jl} P E_k^\dagger E_l P \\
&= \sum_{k,l} u_{ki}^\dagger u_{jl} \alpha_{kl} P \\
&= d_{kl} P
\end{aligned}$$

which simplifies the Knill-Laflamme condition, since d_{kl} is a diagonal matrix. This diagonalization makes it explicit that the error subspaces are orthogonal. We decompose the operation elements as $F_i P = \sqrt{d_{ii}} U_i P$, where U_i is a unitary operator and $\sqrt{d_{ii}}$ is a positive real number.

F_i maps the code subspace C to the error subspace defined by the projector $P_i = U_i P U_i^\dagger = F_i P U_i^\dagger / \sqrt{d_{ii}}$. As we have already discussed, the syndrome measurement is the projective measurement defined by the projectors P_i . The recovery operation $\mathcal{R}$ is defined as $\mathcal{R}(\rho) = \sum_k U_i^\dagger P_i \rho P_i U_i$. For any state ρ supported on C , we can show that:

$$\begin{aligned}
U_i^\dagger P_i F_k \sqrt{\rho} &= U_i^\dagger P_i^\dagger F_j P \sqrt{\rho} \\
&= \frac{U_i^\dagger U_i P F_i^\dagger F_j P \sqrt{\rho}}{\sqrt{d_{ii}}} \\
&= \delta_{ij} \sqrt{d_{ij}} \sqrt{\rho}
\end{aligned}$$

Thus, one can easily check that this recovery operation satisfies the error correction condition:

$$\begin{aligned}
(\mathcal{R} \circ \mathcal{E})(\rho) &= \sum_{i,j} U_i^\dagger P_i F_j \rho F_j^\dagger P_i U_i \\
&= \sum_{i,j} \delta_{ij} d_{ii} \rho \propto \rho
\end{aligned}$$

($\Rightarrow$) Assume $\{E_i\}$ are errors correctable by $\mathcal{R}$, with operation elements $\{R_j\}$. We define a quantum operation $\mathcal{E}_C$ by $\mathcal{E}_C(\rho) = \mathcal{E}(P\rho P)$. $P\rho P \in C$ for any ρ , so we can apply the error correction condition to obtain:

$$\mathcal{R}(\mathcal{E}_C(\rho)) \propto P\rho P$$

We can rewrite the left-hand side of this equation as:

$$\mathcal{R}(\mathcal{E}_C(\rho)) = \sum_{i,j} R_j E_i P \rho P E_i^\dagger R_j^\dagger$$

Thus

$$\sum_{i,j} R_j E_i P \rho P E_i^\dagger R_j^\dagger = c P \rho P$$

for some constant c . The quantum operation with operation elements $\{R_j E_i\}$ is identical to the quantum operation with only one operation element $\sqrt{c}P$. This implies that there exist a complex matrix with entries c_{ki} such that $R_k E_i P = c_{ki} P$. We take the adjoint and obtain $P E_i^\dagger R_k^\dagger R_k E_j P = c_{ki}^* c_{kj} P$. Since $\mathcal{R}$ is a CPTP map, we have $\sum_k R_k^\dagger R_k = I$. Thus, we can sum over k to obtain:

$$P E_i^\dagger E_j P = \alpha_{ij} P$$

Which is the Knill-Laflamme condition, with $\alpha_{ij} = \sum_k c_{ki}^* c_{kj}$. □

The proof of the Knill-Laflamme condition explicitly provides a method to construct the recovery operation $\mathcal{R}$, given the code subspace and the noise model. However, a quantum system could be affected by a noise process that is not known. It would be desirable to have a code C and an error-correction operation $\mathcal{R}$ that can correct the effect of an entire class of noise processes. That is indeed possible, and it is a consequence of the linearity of quantum mechanics. If we have a code that can correct the effect of a set of errors $\{E_i\}$, then it can also correct the effect of any error that can be expressed as a linear combination of those errors. This is a direct consequence of theorem 1, since if C satisfies the Knill-Laflamme condition for $\{E_i\}$, then it also satisfies it for any linear combination of those errors.

1.3 Stabilizer Codes

The previous discussion has proven the theoretical possibility of quantum error correction. However, the state-vector representation of quantum states carries a practical impossibility. When we have a large number n of physical qubits, the manipulation of the 2^n -dimensional state vector becomes unfeasible. We need a more efficient way to represent quantum states, and the stabilizer formalism provides us with such a tool.

1.3.1 Stabilizer Formalism

If we take a closer look at the Shor code, and specifically at the syndrome measurement, we can see that in order to detect a bit flip on one of the first three qubits, we measured the eigenvalues of the operators $Z_1 Z_2$ and $Z_2 Z_3$. The same procedure should be performed for the other two blocks of three qubits. In order to detect phase-flip errors, we need to measure the eigenvalues of the operators $X_1 X_2 X_3 X_4 X_5 X_6$ and $X_4 X_5 X_6 X_7 X_8 X_9$. In total, we need to measure the eigenvalues of eight operators. The two valid codewords, $|0_L\rangle$ and $|1_L\rangle$, are the eigenstates of all these operators with eigenvalue $+1$, and hence are simultaneously fixed by any of them. Those are called stabilizer operators, and the basic idea of the formalism is to describe the code space as the subspace stabilized by these operators.

In order to give a more precise description of the stabilizer formalism, let's introduce the n -dimensional Pauli group $\mathcal{P}_n$, which is the group generated by the n -fold tensor products of the Pauli matrices and the identity, together with the multiplicative factors ± 1 and $\pm i$. For instance, the 1-dimensional Pauli group is

$$\mathcal{P}_1 = \{\pm I, \pm X, \pm Y, \pm Z, \pm iI, \pm iX, \pm iY, \pm iZ\}$$

It is easily verifiable that $\mathcal{P}_n$ is a group under matrix multiplication. The stabilizer is some Abelian subgroup $\mathcal{S} \subseteq \mathcal{P}_n$ that does not contain $-I$, and the vectors fixed by all elements of $\mathcal{S}$ form the code space C , i.e.:

$$C \equiv \{|\psi\rangle \in \mathcal{H} : g|\psi\rangle = |\psi\rangle; \forall g \in \mathcal{S}\}$$

It is also obvious that the code space C is indeed a vector space. $\mathcal{S}$ is necessarily an Abelian subgroup, since two non-commuting operators cannot have common eigenstates.

Let's recall that our aim is to define a subspace C of the total Hilbert space, such that it encodes k logical qubits into n physical qubits. Therefore, the dimension of C must be 2^k . We want to see how this condition translates on the definition of the stabilizer $\mathcal{S}$:

Proposition 1. *If $\mathcal{S}$ is an Abelian subgroup of $\mathcal{P}_n$ that does not contain $-I$, then the code space C stabilized by $\mathcal{S}$ has dimension 2^{n-k} , where k is the number of independent generators of $\mathcal{S}$.*

Proof. Let $\mathcal{S}$ be generated by $\langle g_1, g_2, \dots, g_k \rangle$. Since all the generators are elements of $\mathcal{P}_n$, they are unitary and Hermitian, and thus have eigenvalues ± 1 . We can project the total Hilbert space into the $+1$ eigenspace of each generator g_i by applying the projector

$$P_i = \frac{I + g_i}{2}$$

The code space C is the subspace stabilized by all the generators, and thus can be obtained by applying the projectors associated with each generator to the total Hilbert space:

$$P_C = \prod_{i=1}^k P_i = \prod_{i=1}^k \frac{I + g_i}{2}$$

If we expand the product, we sum over all possible products of the generators g_i , which yields the 2^k elements of $\mathcal{S}$.

$$P_C = \frac{1}{2^k} \sum_{g \in \mathcal{S}} g$$

The dimension of the code space C is given by the trace of the projector P_C :

$$\dim(C) = \text{Tr}(P_C) = \frac{1}{2^k} \sum_{g \in \mathcal{S}} \text{Tr}(g)$$

We have the following cases for the trace of an element $g \in \mathcal{S}$:

- If $g = I^{\otimes n}$, then $\text{Tr}(g) = 2^n$.
- If $g \neq I^{\otimes n}$, then g is a non-trivial element of $\mathcal{P}_n$, and thus has trace zero, since it contains at least one Pauli matrix that has zero trace.

$-I \notin \mathcal{S}$ ensures that there are no elements of $\mathcal{S}$ with trace -2^n . Thus, the only element of $\mathcal{S}$ with non-zero trace is the identity, and we have:

$$\dim(C) = \text{Tr}(P_C) = \frac{1}{2^k} \cdot 2^n = 2^{n-k}$$

□

Therefore, a code that encodes k logical qubits into n physical qubits can be defined by a stabilizer $\mathcal{S}$ with $n - k$ independent generators. This proposition also provides a reason for the non-inclusion of $-I$ in the stabilizer. This is nonetheless obvious from the fact that the only solution to $-I|\psi\rangle = |\psi\rangle$ is the trivial solution $|\psi\rangle = 0$.

We have discussed the description of a subspace using stabilizer formalism. Now let's see how we can describe the dynamics of those subspaces under the action of noise and other quantum processes. Let $|\psi\rangle$ be an element of a code space C stabilized by $\mathcal{S}$. If we apply an unitary operation U ,

$$U|\psi\rangle = Ug|\psi\rangle = (UgU^\dagger)U|\psi\rangle$$

and we see that $U|\psi\rangle$ is stabilized by the conjugated operator $UgU^\dagger$. Since this is common to any element of C , we can easily see that UC is stabilized by $USU^\dagger \equiv \{UgU^\dagger : g \in \mathcal{S}\}$. This property does not ensure that any stabilizer of a subspace UC is a valid stabilizer, since the conjugation by U might introduce elements that are not in $\mathcal{P}_n$. We hence introduce the Clifford group $\mathcal{C}_n$ as the stabilizer of the Pauli group:

$$\mathcal{C}_n \equiv \{U : U\mathcal{P}_nU^\dagger = \mathcal{P}_n\}$$

The Clifford group is the largest group of unitary operations that preserve the Pauli group under conjugation. If we apply an operation $U \in \mathcal{C}_n$ to a code space C stabilized by $\mathcal{S}$, the resulting subspace UC is stabilized by $USU^\dagger \in \mathcal{P}_n$. Conjugation by a Clifford operation also preserves the commutation relations

$$Ug_1U^\dagger Ug_2U^\dagger = Ug_1g_2U^\dagger = Ug_2g_1U^\dagger = Ug_2U^\dagger Ug_1U^\dagger$$

which ensures that the resulting stabilizer $USU^\dagger$ is also an Abelian subgroup of $\mathcal{P}_n$. If $-I$ was in $USU^\dagger$, then there would be some $g \in \mathcal{S}$ such that $UgU^\dagger = -I$, and therefore:

$$\begin{aligned} U^\dagger(UgU^\dagger)U &= U^\dagger(-I)U \\ g &= -I \end{aligned}$$

which contradicts the definition of stabilizer. Thus, $-I$ is not in $USU^\dagger$, and the resulting subspace UC is a valid code space stabilized by $USU^\dagger$.